Youssef Chouay

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Education

Masters in Computer Science - Thesis Ottawa, ON
University of Ottawa *Jan 2025 – Dec 2026*
- **Supervisor:** Vida Dujmovic (Wikipedia Link) | Received over **$52,000** in research scholarships

Bachelor of Applied Science in Software Engineering Ottawa, ON
University of Ottawa *Sept 2020 – Dec 2024*
- **Relevant Coursework:** Data Structures & Algorithms, Embedded Systems, Databases, Discrete Math, Real-Time Systems Design, Enterprise Architecture

Experience

Software Engineer Intern Jan 2026 – Present
TrendMicro/TrendAI *Ottawa, ON*
- Owned end-to-end delivery of a backend service for the CRQ service to fetch company profiles and stock data via **MCP**-based integrations, including research, orchestration design, implementation, testing, and production deployment on **AWS**.
- Led migration of an MLOps codebase to a **Lakehouse** architecture by rewriting data pipelines in **PySpark**, helping enable scalable processing and improved maintainability of analytics workflows.

Artificial Intelligence Researcher May 2024 – Present
National Research Council *Ottawa, ON*

AI-Enhanced Building Automation (BAS) for Modern Facilities:
- Published, as **First-author**, a peer-reviewed paper called "**Implementing AI in Smart Buildings: A Modular, Proof-of-Concept approach**" and presented to field experts at **IEEE EPEC 2025**
- Designed and deployed using **Python/LangChain** agents bridging BAS and LLM tools, cutting data-processing time and operator workload by **49%**.
- Built a **SQLite**-backed ingestion pipeline to process and integrate real-time BAS streams reliably.
- Partnered with **Delta Controls** and **Carleton University** to deliver AI building agents, achieving a **56%** reduction in maintenance costs via automated fault detection, predictive maintenance, and real-time alerts.

Enhanced Utilities chatbot:
- Built a multi-agent, tool-using utilities chatbot that explains bills, simulates alternate rate plans, and diagnoses anomalies by linking AMI data with **weather/holidays/tariffs** for a **$3M+** annual revenue company.
- Implemented a time-series engine (seasonal decomposition + change-point detection) to flag **spikes, persistent baseload, and overnight leaks**, then auto-generate plain-English "why it happened" narratives and savings playbooks.

Software Engineer Intern Sept 2022 – Aug 2023
Wind River Systems *Ottawa, ON*
- Designed and implemented an Automation Dashboard using **Angular**, **TypeScript**, and **Django**, with a **PostgreSQL** database, to streamline the management and analysis of services used by industry leaders such as **NASA, Airbus, and Ford**.
- Achieved over **90%** faster query execution and UI responsiveness by optimizing API endpoints, implementing efficient database queries, and reducing frontend rendering times.

Software Developer May 2022 – Apr 2024
University of Ottawa *Ottawa, ON*
- Redesigned and optimized the university's search engine using **PHP**, **MySQL**, and **Apache**, improving query response times by **80%**, benefiting over **5,000+ students** and saving the university over **$30,000** annually.
- Developed and deployed automation scripts using **PHP**, **Bash**, and **Cron jobs** to enhance search speed by **54%** and streamline data migration workflows.

Teaching Assistant Sept 2023 – Present
University of Ottawa *Ottawa, ON*
- Assisted in teaching **Graduate** classes such as **Machine Learning for Bio-informatics** and **Undergraduate** ones like **Data Structures & Algorithms**, **Design & Analysis of Algorithms**, **Programming Paradigms** and **Discrete Structures**.

Projects

AtlasLLM| *Personal Project*
- Built a **multi-model AI chat platform** with a using **Next.js 14, Convex, Cloudflare buckets and Redis**, supporting **30+ LLMs** across 9 providers through a unified OpenRouter abstraction, generating over **$3,000 in profit**.
- Engineered a **custom DAG-based workflow orchestration engine** powering agentic research pipelines (deep research, pro search) with parallel task fan-out, typed event-driven state, retry/timeout policies, and **Langfuse** tracing per generation.
- Integrated **Clerk** for authentication and subscription billing (backed by **Stripe**), **atomic Redis Lua scripts** for thread-safe credit deduction, and a production observability stack spanning **Sentry, PostHog, Axiom, and OpenTelemetry** for error tracking, product analytics, and distributed tracing.

GeeGee's Intramural website | *Personal Project*
- Built a **GeeGees Intramural Sports Hub** from scratch using Next.js + TypeScript delivering an accessible, responsive experience for thousands of students.
- Designed a **high-throughput Rust + Actix-web API** backed by **SQLx/PostgreSQL** that streams real-time standings, Elo ratings, and predictive match analytics with sub 20ms latency.
- Drove concurrency with async/await and strict type-safety to create a modular, fault-tolerant codebase that scales gracefully under heavy traffic.